23AID498 – Project Phase 1

Project Diary

Student Name:	Rushik Rohan Chitta
Roll No:	AM.AI.U4AID23035
Team No:	13
Guide Name:	Akshara P Byju
Project Title:	An Enhanced BROIE framework for Semi-Supervised ROI Extraction using Multispectral Fusion and Advanced Attention with Geospatial Localisation.

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 9/07/2026

Work planned for the coming week	• Search and collect recent research papers. • Study the selected papers to understand methodologies, datasets and evaluation techniques. • Prepare a comparative literature review.
Work completed during the previous week	• Finalized the project title, objectives and problem statement. • Identified suitable datasets • Discussed the project proposal with guide and received approval to proceed with literature review.
Pending work	• Complete the literature survey and finalize the most relevant research paper. • Identify the baseline model and implementation approach. • Prepare the literature review report and begin implementation phase.

Remarks by the Guide: Mel last week

Weekly Performance Rating (0–5): 5 /5

Guide's Signature with Date: 13/7/2026.

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 25-7-2026

Work planned for the coming week	→ Design system architecture and finalize the methodody ~~from~~ By filling the Research gaps
Work completed during the previous week	⇒ Find 5 Paper each for literature review and make a Presentation for zeroth Presentation regarding this initial research Paper
Pending work	→ finalize the System Design

Remarks by the Guide: Literature survey conclusion should be re-worked. No proper justification between ideas and conclusions

Weekly Performance Rating (0–5): 3 / 5

Guide's Signature with Date:

27-Jul-26

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 3/8/2026

Work planned for the coming week	Implementation of Phase1 System Design and validate it Properly. → Set up the required environment for implementation
Work completed during the previous week	Design the system Artitecture and improvise its workflow → Finalize the datasets → finalize model Pipeline.
Pending work	→ Complete the Phase 1 implementation → And evaluate the trained model.

Remarks by the Guide: Proceed with implementation

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: 3.8.26

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 18/08/26

Work planned for the coming week	• Complete the Proper Training and model convergence • Generate segmentation masks • Proceed with geospatial analysis
Work completed during the previous week	• Completed the PrePrcessing and initiated the training of complete dataset. • Completed the Training Off the dataset.
Pending work	• Complete remaining training epochs & finetuning • Generate visulizations • Perform GeoSpatial analysis

Remarks by the Guide: Proceed

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Akshara P. Bijo

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 28/08/26

Work planned for the coming week	* Complete the full training epoches and model convergence. * Generate segmentation masks. * Proceed with geospatial analysis.
Work completed during the previous week	* Completed training of the Semantic Segmentation model on the Vaihigen dataset * performed validation using the preprocessed validation Patches. * Obtained validation loss and MIOU results
Pending work	* Completed the remaining training epoches & fine tuning * Generate & compare mask for difficult classes. * Perform Geospatial analysis.

Remarks by the Guide: Should improve the technical capability.

Weekly Performance Rating (0–5): 3 / 5

Guide's Signature with Date: Akshara P Biju

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 19/9/26

Work planned for the coming week	implement the proposed model on the dataset on paper • Add more novelty & reconstruct the architecture diagram based on the enhanced work.
Work completed during the previous week	• completed the mid review & incorporated the guide feedback • implemented & evaluated the ~~baseline~~ enhanced BROIE model our dataset. (Coltc2020)
Pending work	• complete implementation & testing of the enhanced model on the ~~paper~~ Previous Paper dataset (RGB) (WHU) • Finalize the novelty & reconstructed architecture, followed by comparison

Remarks by the Guide:

Very good. Proceed.

Weekly Performance Rating (0–5): 4 / 5

Guide's Signature with Date:

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 8/10/26

Work planned for the coming week	• implement the enhanced BOOIE + ORPL model on WHU dataset & evaluate its performance • Prepare the Phase 2 Project Plan.
Work completed during the previous week	• Completed the code review. & got the changes from guide • Finalized ORPL as the core novelty & Prepare the system architecture diagram
Pending work	• Run & evaluate the NAV pipeline on our dataset (DFC2020) • Finalize Phase 2 Plan

Remarks by the Guide:

Code review completed

Weekly Performance Rating (0–5): 4 / 5

Guide's Signature with Date:

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 06/11/2026

Work planned for the coming week	• Implement the architecture on another dataset • Analyze & compare the results. • Work on project report.
Work completed during the previous week	• Completed the proposed architecture implementation. • Worked on DFC2020 with 50:50, 30:70 & 10:90 data splits. • Completed training & evaluation.
Pending work	• Implementation on another dataset. • Cross-data performance comparison • Project report completion.

Remarks by the Guide: Should the result on one data set

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: 6/11/26 Akshara P. Bijo

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 11/11/26

Work planned for the coming week	• Finish the Report • Prepare for final presentation • PPT for submission
Work completed during the previous week	• worked on second dataset with & without ORPL (Postdam) • worked on first dataset with & without ORPL (vaihingen) • completed Phase-1
Pending work	• Report submission final draft • PPT preparation

Remarks by the Guide:

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date:

Akshara P Bijoy

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 12/11/26

Work planned for the coming week	• Complete and prepare for final presentation.
Work completed during the previous week	• Finished Phase I • Changed architecture diagram according to guide's guidelines.
Pending work	• Prepare for final presentation.

Remarks by the Guide:

Weekly Performance Rating (0–5): 4 / 5

Guide's Signature with Date:

Akshara P Bijo